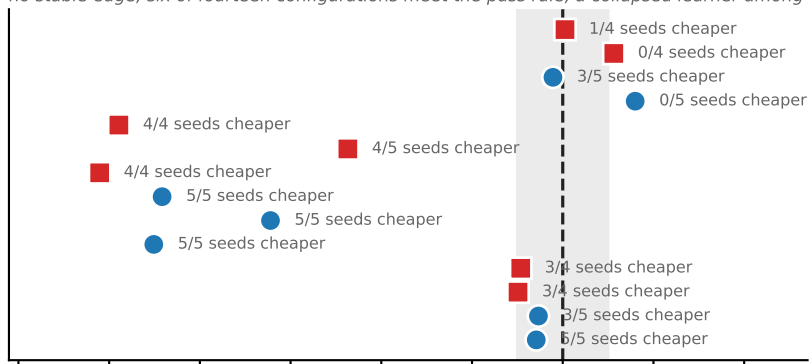


Eighteen pre-registered configurations, by environment

1. Recorded order books

no stable edge; six of fourteen configurations meet the pass rule, a collapsed learner among them

1-min bars, 30-min · DQN 96.57 BTC
 1-min bars, 30-min · DQN 193.13 BTC
 1-min bars, 30-min · PPO 96.57 BTC
 1-min bars, 30-min · PPO 193.13 BTC
 10-s bars, 30-min · DQN 96.57 BTC
 10-s bars, 30-min · DQN 193.13 BTC
 10-s bars, 30-min · PPO 386.27 BTC
 10-s bars, 30-min · PPO 96.57 BTC
 10-s bars, 30-min · PPO 193.13 BTC
 10-s bars, 30-min · PPO 386.27 BTC
 10-s bars, 10-min · DQN 193.13 BTC
 10-s bars, 10-min · PPO 96.57 BTC
 10-s bars, 10-min · PPO 193.13 BTC



2. Reacting simulator, no injected signal

null — both sealed confirmations fail

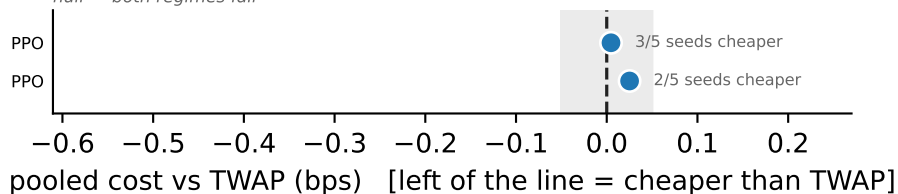
selected config (lr 1e-3), volatile · PPO
 remedial pick (bigger net), volatile · PPO



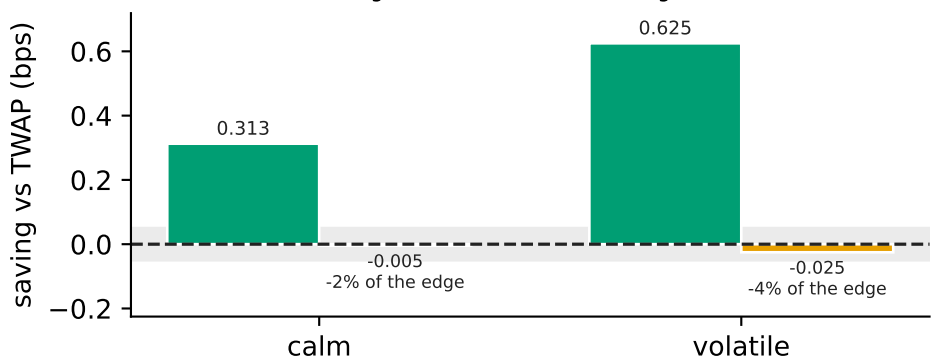
3. Reacting simulator with the measured signal

null — both regimes fail

calm regime · PPO
 volatile regime · PPO



Environment 3: the edge was available and the agents left it



● PPO configuration --- TWAP ■ registered rule, zero fitted parameters
 ■ DQN configuration ■ +/-0.05 bps materiality band ■ trained agents